Fiscal Year:	FY 2026	**Task Last Updated:**	FY 04/07/2026
PI Name:	Sackett, Charles Ph.D.		
Project Title:	AS-PHYS-Bloch Oscillations in Microgravity LOI		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sackett@virginia.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	434-924-6795
Organization Name:	University of Virginia		
PI Address 1:	Physics		
PI Address 2:	382 McCormick Rd		
PI Web Page:			
City:	Charlottesville	**State:**	VA
Zip Code:	22904-1000	**Congressional District:**	5
Comments:	Other names: CA Sackett; Cass Sackett.		
Project Type:	Flight	**Solicitation / Funding Source:**	2022 Physical Sciences NNH22ZDA001N-FP E.6 Fundamental Physics
Start Date:	06/05/2024	**End Date:**	09/30/2029
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:	Walsh, Daniel	**Contact Phone:**	
Contact Email:	daniel.h.walsh@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 09/30/2029 per J. Callas/JPL (Ed., 6/24/25). NOTE: End date changed to 09/30/2028 per D. Griffin/NASA-HQ. The original period of performance was 06/05/24 - 06/04/28 (Ed., 1/21/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC24K1237		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

The upgrade of the NASA Cold Atom Laboratory (CAL) provides the opportunity for new flight experiments. We propose here to explore the phenomenon of Bragg oscillation in a microgravity environment, with applications to precise and robust measurements of small accelerations at the level of 1 cm/s^2. A parallel set of ground experiments will help guide the flight effort.
Bloch oscillations occur when atoms are held in an optical lattice potential, in the presence of an accelerating force pointing along the lattice direction. The force causes the atoms to tunnel through the lattice barriers, so the atoms accelerate much as they would in free space. However, when the velocity of the atoms reaches the Bragg velocity of the lattice, the atoms are coherently reflected and their momentum is reversed. The accelerating force continues to act and the momentum increases again, with the cycle repeating as long as the atoms can be retained. Bloch oscillations are an interesting physical phenomenon with many applications. For example, by measuring the period of the oscillation, the strength of the acceleration can be determined with high precision [1]. [Ed. Note: See References.]

In terrestrial experiments, the acceleration is typically due to gravity and the amplitude of the oscillation is on the order of micrometers, since the atoms do not fall very far before reaching the Bragg velocity of order 1 cm/s. In microgravity, the oscillation amplitude is much larger, reaching the mm scale at milli-g accelerations. A few terrestrial experiments have investigated Bloch oscillations at low accelerations using horizontal optical lattices. However, these efforts are hampered by the need to support the atoms against gravity. This requires a combination of a relatively tight optical trap and magnetic levitation. The tight trap leads to high atomic density and thus complicating effects from interactions. Magnetic levitation cannot easily be made uniform, leading to spatial variations in the accelerating force. For instance, the experiments by Geiger et al. [2] observed oscillation amplitudes of about 150 um, but required a Feshbach resonance to nullify interactions and were limited by inhomogeneity. The sensitivity to accelerations increases with the amplitude of oscillation, so larger amplitudes are desirable.

The microgravity environment on CAL offers a clear opportunity for improvement. The existing Bragg laser can be used to provide the optical lattice, where a depth of about one recoil energy is needed. A tunable acceleration can be obtained using the magnetic gradient control provided in the new upgrade. A gradient of 15 mG/cm gives a suitable acceleration of 1 cm/s^2. The period of the oscillation would be about 0.5 s, which is within the vacuum limits of the system.

Although Bloch oscillations can be described using classical language, they are in fact a quantum phenomenon; classical particles would not "Bragg reflect" from a lattice. The amplitude of the oscillation is set by the spatial extent of the Wannier-Stark wavefunctions of the lattice, and the atomic waves are coherent across this length scale. Large Bloch oscillations are thus comparable to atom interferometry, but Bloch oscillations can be more robust to technical imperfections. For instance, they do not require a well-defined initial atom velocity, and they can tolerate a range of lattice depths. Our group is well-situated to carry out ground studies to support the flight operations. We have an existing apparatus that provides weakly-confined atoms in a highly uniform horizontal magnetic guide. We propose to use this system to test the approach and technical requirements, so that the flight experiments can be more quickly achieved and optimized.

References

[1] Nalcz I, Masi L, Ferioli G, Petrucciani T, Fattori M, Chwedenczuk J. Sensitivity bounds of a spatial Bloch-oscillation atom interferometer. Physical Review A. 2020 Sep;102(3):033318. https://

[2] Geiger ZA, Fujiwara KM, Singh K, Senaratne R, Rajagopal SV, Lipatov M, Shimasaki T, Driben R, Konotop VV, Meier T, Weld DM. Observation and uses of position-space Bloch oscillations in an ultracold gas. Physical review letters. 2018 May 25;120(21):213201. https://

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The primary impact of the research will be on the development of tools for use in further ultracold atom research in microgravity. In particular, the work will demonstrate improved atom cooling techniques, and it will develop Bloch oscillations as a tool for atom interferometry. These techniques offer unique opportunities in microgravity, but will work differently than in a terrestrial environment. Long-term applications for these techniques include inertial sensing, gravity mapping, and tests of Einstein’s equivalence principle.

Progress on the main tasks of the project has remained limited by hardware problems. However, we have continued to support NASA's Jet Propulsion Laboratory (JPL) in development of the new SM3X module, which is launching in April 2026. This module will restore both atom interferometry and dual-species capability, along with the new meso-chip coils to assist condensate production and field compensation. These will be important enablers for our projects.
One potential issue is an alignment deviation with the Bragg laser beam, which is the laser we plan to use for Bloch oscillation studies. During vibration testing of the module, the optical alignment of this laser shifted slightly so that the beam is no longer perfectly aligned with its retro-reflection. An extensive investigation at JPL indicated that the misalignment is not severe enough to prevent using the beam for atom interferometry or Bloch oscillations, but the in-situ performance will need to be evaluated once the module is operational.

In the meantime, we have continued our work with cold but non-condensed atoms in the currently-installed SM1 module. This effort focuses on atoms confined in a “quadrupole trap” that is formed by opposing pairs of electromagnetic coils. Normally, the Cold Atom Laboratory (CAL) uses this type of trap to transfer atoms from a laser-cooled magneto-optical trap to a magnetic trap formed by an atom chip, where Bose condensates are produced. During the current performance period, electronics faults prevented reliable operation of the chip trap, but the quadrupole trap was not affected. Our goal is to understand the behavior of atoms in the quadrupole trap and optimize its performance so as to improve condensate production in SM3X.

For one set of studies, we have investigated the lifetime of atoms in the quadrupole trap. Atom loss in this system is expected to be dominated by two processes. The first is through collisions with residual room-temperature molecules in the vacuum chamber. These molecules are energetic enough to knock atoms out of the quadrupole trap. The loss rate is set by the background pressure of the CAL science cell. The second loss mechanism is through Majorana transitions. A defining feature of the quadrupole trap is that the trapping magnetic field has a zero at the trap center. Normally the magnetic moment of an atom is locked to the direction of the local magnetic field, which preserves the spin state of the

Task Progress:	atom. But if an atom moves too close to the center, there is no field to lock to and the spin state can spontaneously change. Because only atoms in the correct spin state are confined, this leads to the atom being ejected from the trap. Majorana losses have been measured on the ground, where the lifetime is found to depend on the spatial size of the atom cloud as τ≈(100 s√mm^2) σ^2, where σ is the rms cloud radius. Lifetime measurements on CAL show an unexpected result. For large clouds, we obtain a vacuum-limited lifetime τ_0 of about 15 s, which is on par with expectations. However, for smaller clouds we observe a lifetime that seems to scale more rapidly with the cloud size, as (6×10^4 s√mm^6)σ^6. The reason for this discrepancy is not yet clear, but a potential explanation is that magnetic field noise causes the position of the trap center to move in space, which would increase the volume in which atoms are rapidly lost. If this is correct, efforts to abate the noise could improve CAL performance. We will continue to explore this question until SM3X is installed. In ground-based efforts, we continue to rebuild our atom interferometry apparatus to support studies with atom chips. We will load the chip by transporting atoms from a magnetic quadrupole trap to a glass cell with a thin silicon wall. The atom chip will be located just outside the wall. During the current period, we implemented the optical tweezers and observed transport, but unfortunately an obstruction in the vacuum chamber limited the transport distance. We have repaired this fault and have now observed the required range of motion. The next step is to move the atoms adjacent to the silicon wall. We achieve this by displacing the tweezer beam with a tilting mirror stage, but we have found that activating the tilt stage causes significant loss of atoms. We are currently investigating the reason for this and exploring other options. We have also developed procedures to produce our own atom chips using photolithographic techniques. We will explore a novel chip geometry based on two wires in a cross pattern, in conjunction with bias fields parallel to the chip plane. This geometry can provide both three-dimensional traps and two-dimensional wave guides, making it potentially a highly flexible tool for atom interferometry applications. Chips and drivers based on this approach have been implemented and tested. Finally, we have been doing theoretical work to better understand the noise performance of differential atom interferometry. This is a common technique where two interferometers are implemented at the same time, such that the effect of interest (for instance, a rotation or a gravity gradient) has a different effect on each, while other noise sources (for instance, vibrations or laser noise) have the same effect. By observing the phase difference between the two interferometer outputs, the differential effect can be extracted even in the presence of substantial noise. While several techniques are available for carrying out this extraction, less is known about the impact of noise sources on the precision that can be achieved. We have numerically compared the performance of several methods, and in the course of this study developed a new technique that is advantageous in some situations. We are currently preparing this work for publication.
Bibliography Type:	Description: (Last Updated: 04/22/2026)
Abstracts for Journals and Proceedings	Chu Z, Sackett CA. "Quantum rotation sensing by BEC in an atom chip waveguide." 56th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics (DAMOP 2025), Portland, OR, June 16-20, 2025. Bulletin of the American Physical Society. 56th Annual Meeting of the APS Division of Atomic, Molecular and Optical Physics (DAMOP 2025), Portland, OR, June 16-20, 2025. Abstract D00.00145. https://meetings.aps.org/Meeting/DAMOP25/Session/D00 , Jun-2025
Abstracts for Journals and Proceedings	De Urioste Terrazas I, Sackett CA, Chu Z. "Fabrication and modeling of a cross-trap atom chip for interferometric applications." 92nd Annual Meeting of the Southeastern Section of the American Physical Society (SESAPS 2025), Harrisonburg, Virginia, October 23-25, 2025. Abstracts. 92nd Annual Meeting of the Southeastern Section of the American Physical Society (SESAPS 2025), Harrisonburg, Virginia, October 23-25, 2025. , Oct-2025
Abstracts for Journals and Proceedings	Sackett CA. "Magnetic trapping of cold atoms in microgravity." 41st Annual Meeting of the American Society for Gravitational and Space Research, Phoenix, Arizona, December 3-6, 2025. Abstracts. 41st Annual Meeting of the American Society for Gravitational and Space Research, Phoenix, Arizona, December 3-6, 2025. , Dec-2025